Micro III Notes

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General Equilibrium

Def "what resources available?"

1. L goods, $l = 1, ..., L$ (finite).

i. consumption: apples, leisure time, location, etc.
ii. inputs: labor, land, capital
iii. dated goods: apple today vs apple tomorrow
iv. contingent goods: umbrella w rain vs umbrella w/o rain

2. Initial (total) Endowment: $\omega = (\omega_1, ..., \omega_L) \in \mathbb{R}^L_+$
3. Production

ii. Production set $Y \subseteq \mathbb{R}_L$.
iii. Call $y \in \mathbb{R}^L$ a "feasible production plan" if $y \in Y$.

4. Feasibility Set/Resource Set: $R = \{x \in \mathbb{R}^L : x = \omega + y \text{ for some } y \in Y\}$.

Def "Properties on Y"

1. Y is closed and nonempty
2. Possible to do nothing: $(0, ..., 0) = \vec{0} \in Y$
3. No free lunch: $Y \cap \mathbb{R}^L_+ = \emptyset$
4. Free disposal: $y \in Y, \hat{y} \leq y \Rightarrow \hat{y} \in Y$

Def "feasible"

x is *feasible* if:

1. $x_i \in X_i, \forall i = 1, ..., I$ and
2. $\sum_{i=1}^I x_i = R$ (i.e., $\exists y \in Y : \sum_{i=1}^I x_i = \omega + y$)

Def "Pareto Efficient"

A feasible consumption plan, x is *Pareto Efficient* (Pareto Optimal) if there does not exist any other feasible $\hat{x}$ such that

1. $\hat{x_i} \succsim_i x_i \forall i = 1, 2, ..., I$ and
2. $\hat{x_i} \succ_i x_i$ for at least one $i \in \{1, ..., I\}$.

If such $\hat{x}$ exists, we say $\hat{x}$ *Pareto-dominates* x.

Remark

1. firms are not people (agents).
2. Pareto efficiency is not about equity.

Def "algorithm"

Steps:

1. compute R from ω and Y to figure out feasible consumption plans.
2. $U_i = \{u_i(x_i) : x \text{ is feasible}\}, \forall i = 1, ..., I.$ (set of all possible utility levels)
3. Fix $\bar{u_i} \in U_i, i = 2, ..., I.$

a. $\max_x u_1(x_1)$ subject to $u_i(x_i) \geq \bar{u_i}, i = 2, ..., I.$

4. Change values: $u_i \in U_i, i = 2, ..., I$ and repeat step 3.

Remark: the algorithm is not sufficient UNLESS

1. u_i is continuous and strongly monotone, and
2. $Y = \{y \in \mathbb{R}^L : \vec{y} \leq \vec{0}\}.$

Prop "Pareto $\Rightarrow x^*$ solution"

If x^* is Pareto efficient, then x^* solves step 3 above for some $\bar{u_2}, ..., \bar{u_I}, \bar{u_i} \in U_i \forall i = 2, ..., I.$

Walrasian Equilibrium

Def "Primitives"

1. Individual firms
   - $j = 1, ..., J, Y_j \in \mathbb{R}^L$ such that
   - $$Y = \sum_{j=1}^{J} Y_j = \{y \in \mathbb{R}^L : \exists y_j \in Y_j, \forall j = 1, ..., J \text{ s.t. } y = \sum_{j=1}^{J} y_j\}$$
2. Individual endowment
   - $\omega_i \in \mathbb{R}^L_+, \forall i = 1, ..., I$ such that $\omega = \sum_{i=1}^{I} \omega_i$ (previously we had only ω and not ω_i)
3. Consumer i share in firm j's profits
   - $\underbrace{\theta_i^j \in [0, 1]}_{\text{share of firm j's profits to consumer i for consumption}}$ $, \forall j$ such that $\sum_{i=1}^{I} \theta_i^j = 1$ (consumers own 100% of firms)

Def "Allocation"

An allocation is

$$(\underbrace{x_1, ..., x_I}_{\text{consumption plans}} , \underbrace{y_1, ..., y_J}_{\text{production plans}}) \subseteq \mathbb{R}^{IL+JL}, \quad \underbrace{\forall i, x_i \in X_i}_{\text{consumption is feasible}} , \underbrace{\forall j, y_j \in Y_j}_{\text{production is feasible}} .$$

Def "Walrasian Equilibrium"

$p^*, (x^*, y^*)$ are a Walrasian Equilibrium if the following are satisfied:

1. Profit Maximization: $p^* \cdot y_j^* \geq p^* \cdot y_j, \forall y_j \in Y_j, \forall j = 1, ..., J$
2. Preference maximization: $\underbrace{p^* \cdot x_i^*}_{\text{expenditure}} \leq \underbrace{p^* \cdot w_i}_{\text{endowment income}} + \underbrace{\sum_{j=1}^{J} \theta_i^j (p^* \cdot y_{j,l}^*)}_{\text{income from profits}} \ \forall y_j \in Y_j, \forall i = 1, ..., I$
3. Market clearing: $\sum_{i=1}^{I} x_{i,l}^* = \sum_{i=1}^{I} \omega_{i,l}^* + \sum_{j=1}^{J} y_{j,l}^*, \forall l = 1, ..., l.$

Obs 1 "normalizing prices"

If (p^*, x^*, y^*) is a Walrasian Equilibrium, then $(\lambda p^*, x^*, y^*)$ is a Walrasian Equilibrium $\forall \lambda > 0$

Obs 2 "Walras' law"

If $\succsim_i$ is locally nonsatiated and (p^*, x^*, y^*) satisfies profit maximization then

$$\underbrace{p^* \cdot \sum_{i=1}^{I} x_i^*}_{\text{agg demand}} = \underbrace{p^* \cdot \sum_{i=1}^{I} \omega_i + p^* \cdot \sum_{j=1}^{J} \overbrace{\sum_{i=1}^{I} \theta_i^j}^{=1} (y_j^*)}_{\text{agg supply}} .$$

Obs 3 "markets clear for all but one, then markets clear for all"

If 1. $\succsim_i$ is locally nonsatiated, 2. (p^*, x^*, y^*) satisfies profit maximization, 3. $p_L^* \neq 0$, and 4. $\sum_{i=1}^{I} x_{i,l}^* = \sum_{i=1}^{I} \omega_{i,l} + \sum_{j=1}^{J} (y_{j,l}^*), \forall l = 1, ..., L-1$,

Then $\sum_{i=1}^{I} x_{i,L}^* = \sum_{i=1}^{I} \omega_{i,L} + \sum_{j=1}^{J} (y_{j,L}^*)$

In words: if market clearing holds for all other goods, and the price of the last good is nonzero, then I have market clearing for the last good.

Thm "First Welfare Theorem (WE $\Rightarrow$ Pareto-efficient)"

Suppose $\succsim_i$ is locally nonsatiated $\forall i = 1, ..., I$ and (p^*, x^*, y^*) is a Walrasian Equilibrium. Then (x^*, y^*) is Pareto-efficient.

Remark: "Implicit assumptions First Welfare Thm"

1. firms are price-taking
2. no externalities
3. markets are complete (less relevant to this course, but an important assumption in finance theory.)

Def "system of transfers"

A system of transfers, T is a vector $T = (T_1, ..., T_I) \in \mathbb{R}^I$ such that $\sum_{i=1}^{I} T_i = 0$.

Note:

- $T_i > 0 \Rightarrow$ "receiving transfer"
- $T_i < 0 \Rightarrow$ "paying transfer (tax)"

Def "WE with transfers"

(p^*, x^*, y^*) is a Walrasian equilibrium with transfers T if it satisfies 1. Profit maximization (no change here), 2. Preference maximization: x_i^* maximizes $\succsim_i$ (represented by u_i subject to $p^* \cdot x_i^* \leq p^* \omega_i + \sum_{j=1}^{J} \theta_i^j p^* \cdot y_j^* + T_i$), and 3. Market clearing (no change here either).

Note: we always add T_i even if $T_i < 0$.

Thm "First Welfare Theorem with transfers"

Suppose T is a system of transfers, $\succsim_i$ is locally nonsatiated $\forall i = 1, ..., I$ and (p^*, x^*, y^*) is a Walrasian Equilibrium with T. Then (x^*, y^*) is Pareto-efficient.

Thm "Second Welfare Theorem (PE $\Rightarrow$ WE)"

Under assumptions 1-4, if (x^*, y^*) is Pareto-Efficient then there exists a system of transfers T and $p^* \neq \vec{0}$ such that (p^*, x^*, y^*) is a Walrasian Equilibrium with transfers T.

Assumptions: 1. $\forall i, X_i, \succsim_i$ satisfies * convexity: $x_i \succsim x_i' \Rightarrow \lambda x_i + (1-\lambda)x_i' \succsim x_i', \forall \lambda \in (0,1)$, and * something slightly weaker than strict convexity: $\forall x_i, x_i', \hat{x}_i \in X_i : [x_i \succ_i \hat{x}_i, x_i' \succ_i \hat{x}_i] \Rightarrow \lambda x_i + (1-\lambda)x_i' \succ_i \hat{x}_i, \forall \lambda \in (0,1)$.) 2. Y_j is convex $\forall j = 1, ..., J$. 3. $\succsim_i$ is locally nonsatiated 4. $\forall i$ * $\succsim_i$ is continuous, and * $X_i \in \mathbb{R}^L_{++}$. This one is VERY STRONG – means that EVERY individual has a nonzero amount of EVERY good.

Note: assumption 4 is weaker in MWG and other places, but we will always use this version in this class.

Existence of Walrasian Equilibria

Def "set of price vectors"

$\Delta = \{p \in \mathbb{R}^L : \forall l, p_l \geq 0 \text{ and } \sum_{l=1}^L p_l = 1\}$

Def "excess demand function"

$\varphi : \Delta \to \mathbb{R}^L$ such that $\forall p \in \Delta$:

$$\varphi(p) = \sum_{i=1}^{I} x_i^*(p) - \sum_{i=1}^{I} \omega_i - \sum_{j=1}^{J} y_j^*(p)$$

is called the excess demand function.

$\varphi(p) = \vec{0} \Rightarrow$ WE exists (in WE: demand(p) = supply(p) $\iff$ demand(p) - supply $= 0$ $\iff$ excess demand$(p) = 0$). And, under some assumptions we can get the converse:

WE exists $\Rightarrow \varphi(p) = \vec{0}$.

Note "Implicit assumptions of excess demand fn"

1. We can focus on $p \in \Delta$. That is, in WE, $p_l \geq 0 \forall l$ and for at least one good $l, p_l > 0$.
2. There is a unique solution to the profit and preference maximizing problems (i.e., φ is a function, not a correspondence.). This one isn't really a problem though because we can always generalize later and use Kakutani's instead of Brouwer's fixed point theorem.
3. Excess demand for any good is finite (even for \$0-priced goods).

Thm "sufficient conditions on φ"

Suppose $\varphi : \Delta \to \mathbb{R}^L$ is well-defined (i.e., satisfies implicit assumptions 1-3) and satisfies

1. φ is continuous,
2. φ satisfies Walras' law ($p \cdot \varphi(p) = 0, \forall p \in \Delta$), and
3. if $p_l = 0$ for some l then $\varphi(p_l) > 0, \forall p \in \Delta$,

Then, there exists at least one $p \in \Delta$ such that $\varphi(p) = \vec{0}$

Note (exercise): $f(p) = p \iff \varphi(p) = 0$

Thm "sufficient conditions on primitives"

Suppose $\forall i, \forall j$:

1. $X_i \in [0, m]^L$ where m is "large" (i.e, $m = \max_{l=1,...,L}\{\max_{(x,y)}\{x_l\}\} + 1$)
2. $\omega_{i,l} > 0, \forall l, i$
3. $\succsim_i$ is continuous, strongly monotone, and strictly convex, and
4. $\vec{0} \in Y_j, Y_j$ is convex and satisfies $\forall y_j, y_j' : y_j \neq y_j' \Rightarrow \exists \hat{y}_j \in Y_j$ such that $\hat{y}_{j,l} > \lambda \hat{y}_{j,l} + (1-\lambda)\hat{y}_{j,l}', \forall l$.

Then, φ is well-defined and satisfies 1-3 of Thm "sufficient conditions on φ".

Arrow's Impossibility Theorem

Def "setting for Arrow's Impossibility Theorem"

1. Finite set of "alternatives," $A = \{a, b, c, ...\}$
2. Finite set of individuals, $N = \{1, ..., N\}$
3. $\mathcal{R}$ is the set of all rational (and antisymmetric) preferences over A, ruling out indifferences ($R_i \in \mathcal{R}$ is the preference of individual i).
4. A preference profile is $R = (R_1, ...R_n) \in \mathcal{R}^n$

Goal: $f : \mathcal{R}^n \to \mathcal{R}$

Thm "Arrow's Impossibility Theorem"

Assumptions: 1. Universal domain: any ranking is possible (so long as it is complete, transitive, and anti-symmetric) 2. society's preference must be complete, transitive, and anti-symmetric 3. $|A| \geq 3$

Axioms: 1. Pareto: $aR_ib, \forall i \Rightarrow af(R)b$ 2. Independence of irrelevant alternatives: $\forall i[aR_ib \iff a\hat{R}_ib] \Rightarrow [af(R)b \iff af(\hat{R})b]$ 3. No dictator: $\nexists i \in N : \forall R \in \mathcal{R}^n, f(R) = R_i$

There is no $f : \mathcal{R}^n \to R$ that satisfies all three axioms at the same time.

Def "winning coalition"

Denote $\mathcal{P}(N)$ to be the set of all nonempty subsets of N ($2^N \setminus \{\emptyset\}$).

We say $\mathcal{G} \subseteq \mathcal{P}(N)$ is a set of winning coalition if

1. $\forall G \subseteq \mathcal{P}(N)$ either $G \in \mathcal{G}$ or $N \setminus G \in \mathcal{G}$, but not both. And
2. $[G \in \mathcal{G}, G' \supseteq G] \Rightarrow G' \in \mathcal{G}$

Def "Generalized majority voting"

$f : \mathcal{R}^n \to R$ is a GMV if there exists a set of winning coalitions $\mathcal{G}$ such that $\forall a, b \in A, \forall R \in \mathcal{R}^n$:

$$af(R)b \iff \{i \in N : aR_ib\} \in \mathcal{G}.$$

Term "Condorcet Winner/Condorcet cycle" the pairwise champion or beats-all winner, is formally called the Condorcet winner. The head-to-head elections need not be done separately; a voter's choice within any given pair can be determined from the ranking.

Some elections may not yield a Condorcet winner because voter preferences may be cyclic—that is, it is possible (but rare) that every candidate has an opponent that defeats them in a two-candidate contest. (This is similar to the game rock paper scissors, where each hand shape wins against one opponent and loses to another one).

Term "Social Welfare Function (SWF)" Function with preferences as inputs and a complete ranking as outputs.

Term "Collective Choice Rule (CCR)" Function with preferences as inputs and a winner as output.

Quasilinear Economy

Term "Quasilinear economy"

$$U(x_i, m_i) = g_i(x_i) + m_i, \forall i \text{ where } g_i \text{ is strictly concave.}$$

Called quasilinear because utility is linear in money but not in the consumption good.

Def "cost function"

$c_j : \mathbb{R}_+ \to \mathbb{R}_+$ is a cost function. $c_j(y_j)$ is the amount of money needed to produce y_j units of consumption good l. c_j must be

1. smooth,
2. increasing, $c_j'(y_j) > 0, \forall y_j > 0$,
3. convex $c_j''(y_j) > 0 \forall y_j > 0$ (i.e., the more you produce, the harder it becomes to produce more.)

Def "primitives on quasilinear economy"

1. Goods:
    1. $l = 1$ is "money," $l = 2$ is consumption good
2. Consumers $i = 1, ..., I$:
    1. $\forall i = 1, ..., I, X_i = \mathbb{R} \times \mathbb{R}_+$ (means negative amounts of money are allowed!)
    2. Initial endowment of $(\omega_i, 0)$.
    3. $u_i(m_i, x_i) = m_i + g_i(x_i)$
        1. money has constant marginal utility and MRS does not depend on m_i.
        2. Assumptions on g_i is smooth and $g_i'(x_i) > 0, g_i''(x_i) < 0, \forall x_i > 0$ (this is concave, but more restrictive - *log* satisfies, but x^2 does not). No need for Inada conditions.
3. Firms $j = 1, ..., J$:
    1. firm j's production set: $Y_j = \{(-z_j, y_j) : y_j \geq 0, z_j \geq c_j(y_j)\}$
        1. $z_j > c_j$ would mean that free disposal is employed
        2. $z_j \geq 0$ - we can write this separately, it's just redundant since c_j maps to/from $\mathbb{R}_+$ and $z_j \geq c_j(y_j)$.
4. Consumer i's share of firm j's profits is as before: θ_i^j

Def "allocation"

A vector $(m, x, z, y) \in \mathbb{R}^I \times \mathbb{R}_+^{I+2J}$ where

1. m_i: money of consumer i
2. x_i: consumption good of consumer i
3. z_j: input of money firm j uses
4. y_j: consumption good firm j produces.

An allocation is *feasible* if

1. Plans are feasible:
    1. $\forall i, (m_i, x_i) \in X_i$
    2. $\forall j, (-z_j, y_j) \in Y_j$
2. Markets clear:
    1. $\sum_{i=1}^I m_i = \sum_{i=1}^I \omega_i - \sum_{j=1}^J z_j$ (money is all used up)
    2. $\sum_{i=1}^I x_i = \sum_{j=1}^J y_j$ (demand = supply)

Def "Walrasian Allocations"

1. Simplifications without loss:
    1. Normalize price of money to 1, price of $l = 2$ to $p > 0$.
        1. always works in a quasilinear economy since utility is strictly increasing in both goods.
        2. focus on the market for the consumption good (since $p > 0$ we apply observation 3)
2. Solve for Demand

UMP is:

$$\max_{m_i \in \mathbb{R}, x_i \in \mathbb{R}_+} m_i + g_i(x_i) \text{ s to } m_i + px_i \leq \omega_i + \sum_{j=1}^{J} \theta_i^j (py_j - c_j(y_j))$$

But we can assume that $m_i + px_i \leq \omega_i + \sum_{j=1}^{J} \theta_i^j (py_j - c_j(y_j))$, isolate, and plug m_i into the objective function to get the unconstrained maximization problem:

$$\max_{x_i \geq 0} \omega_i + \sum_{j=1}^{J} \theta_i^j (py_j - c_j(y_j)) - px_i + g_i(x_i)$$

Which is the same as

$$\max_{x_i \geq 0} g_i(x_i) - px_i$$

So, the UMP for demand in a quasilinear economy is unaffected by money

By FOC and SOCs we get the Econ 101 result that consumers will demand more until MU = MC (unless demand is 0):

$$\begin{cases} MU = g_i'(x_i) = p = MC \text{ if } p \leq g_i'(0) \\ 0, \text{ otherwise} \end{cases}$$

$p \leq g_i'(0)$ since MU is decreasing, so MU is highest at zero

Caution: watch out for corner solutions since x_i can equal 0!

This is why the partial equilibrium from intermediate micro leads to the same conclusions as general equilibrium - we just always studied quasilinear economies in intermediate micro

3. Solve for Supply

Profit max problem:

$$\max_{y_j \geq 0} py_j - c_j(y_j$$

which is the same as

$$\begin{cases} MC = c_j'(y_j) = p \text{ if } p \geq c_j'(0) \\ 0 \text{ otherwise.} \end{cases}$$

The above assumes that MC of production is increasing, so $c_j'(0)$ is the lowest marginal cost

Econ 101: MC = p, unless supply = 0.

Remark: Market supply = sum of all firm's supply

Pareto Efficient Allocations

Allocation of money does not matter

Summing utilities is okay to do in a quasilinear economy - in general it is not!

Prop 1 "PE iff max welfare"

In the quasilinear model a feasible allocation (m, x, z, y) is Pareto Efficient if and only if (x, y) maximize $\sum_{i=1}^{I} g_i(x_i) - \sum_{j=1}^{J} c_j(y_j)$ among all feasible allocations.

That is, $\forall i, j, x_i \geq 0, y_j \geq 0$:

1. $\sum_{i=1}^{I} x_i = \sum_{j=1}^{J} y_j$ and
2. $\sum_{i=1}^{I} m_i = \sum_{i=1}^{I} \omega_i - \sum_{j=1}^{J} z_j$, where $z_j = c_j(y_j)$.

Prop 2 "PE iff max over all feasible allocations"

In the quasilinear model a feasible allocation (m, x, z, y) is Pareto Efficient if and only if (x, y) maximizes the sum of consumer utilities, $\sum_{i=1}^{I}(m_i + g_i(x_i))$, among all feasible allocations.

Econ 101: surplus = sum of utilities of consumption good - sum of firms' costs

Aggregation

Term "summation of sets"

$$M_1 + M_2 = \{x_1 + x_2 : x_1 \in M_1, x_2 \in M_2\}$$

Def "Supply correspondence of firm j"

The supply correspondence of firm j with production set Y_j is S_j, where $\forall p \in \mathbb{R}^L$:

$$S_j(p) = \{y_j \in Y_j : p \cdot y_j \geq p \cdot \hat{y}_j, \forall \hat{y}_j \in Y_j\}$$

- We assume $S_j(p)$ is well defined $\forall p$ – that is, profit maximization has a solution
- In general, $S_j(p)$ is NOT a singleton set.

Def "Aggregate supply correspondence"

$$S^*(p) = (S_1 + \cdots + S_J)(p) = S_1(p) + \cdots + S_J(p)$$

Prop "S looks like the supply correspondence of just one firm"

Denote production set, $Y = Y_1 + \cdots Y_J$ and S is the supply correspondence of a firm with production set Y. Then $S = S_1 + \cdots S_J$.

Def "Setting for demand aggregation"

Prop "demand fns allow aggregation iff"

FINISH

Misc

Def "numeraire"

a tradable economic entity in terms of whose price the relative prices of all other tradables are expressed (i.e., the price that is normalized to [1 or p - need to figure this out!])

w/o free disposal, have to consume everything. w/ free disposal, need strongly monotone preferences to guarantee that consumers consume the entire endowment (budget constraint will bind).